

Appendices

Appendix: Pipeline Stage Reference I

Table 1: Single-project DAG exported from default pipeline.yaml (names shown in topological order). Scripts live under scripts/.

Stage name	Script / method	Primary inputs	Outputs / artefacts	Failure mode
Clean Output Directories	_run_clean_outputs	prior projects/<name>/output/, mirrored output/<name>/ targets	emptied trees	Blocking
Environment Setup	00_setup_environment.py	toolchain probes	scaffold dirs, env exports	Blocking
Infrastructure Tests	01_run_tests.py --infra-only --infra-scope pipeline-smoke	tests/infra_tests	coverage + junit-style logs	tolerant ceilings
Project Tests	01_run_tests.py --project-only	projects/<name>/tests	coverage artefacts	blocking by default
Project Analysis	02_run_analysis.py	thin scripts	figures/, data/, reports	Blocking
PDF Rendering	03_render_pdf.py	manuscript/, placeholders	.pdf/.tex bundles	Blocking
Output Validation	04_validate_output.py	render tree	Markdown + PDF diagnostics	Blocking /
LLM Scientific Review	06_llm_review.py --reviews-only	resolved manuscript artefacts	JSON textual reviews	downgrade
LLM Translations	06_llm_review.py --translations-only	abstract metadata	multilingual snippets	Optional skip (allow_skip)
Copy Outputs	05_copy_outputs.py	validated tree	mirrored output/<name>/...	Optional skip (allow_skip) soft fail logged

Appendix: Pipeline Stage Reference II

Executable	08_executable_bundle	project tree +	container bundle manifest	opt-in (bundle tag)
Bundle	.py	outputs		
Archival	09_archive_publicati	bundle + deliverables	archival deposit manifest	opt-in (archival
Publication	on.py			tag)

Appendix: Pipeline Stage Reference III

`scripts/07_generate_executive_report.py` is invoked **outside** this DAG whenever `execute_multi_project.py` aggregates pipelines—supplying cross-project KPI dashboards absent from lone-project checkpoints.